

Fault Lines in a Fly Brain

Mapping structural resilience in sensory-to-motor pathways across the complete male *Drosophila* central nervous system

A focused, defensible study built around the MaleCNS v1.0 connectome. The paper asks a simple question: which parts of the fly nervous system can lose connections and keep a route from sensation to action, and which parts fail when only a few critical neurons are removed?

166,700

neurons reported in the final peer-reviewed Cell article [1]

11,710

neuron types reported in the final peer-reviewed Cell article [1]

124.2M

proofread synaptic connections reported in the final Cell Results section [1]

Recommended paper type: applied computational neuroscience and network analytics

Primary contribution: a reproducible lesion and redundancy benchmark for structural sensorimotor pathways

Practical fit: no wet lab, no claim of biological behavior, and a clear path to code, figures, and an interactive companion

The candidate question is resilience, not whether a connectome can play a game

The dataset is new, but connectome-shaped controllers already exist. A formal literature review must determine whether the proposed whole-CNS resilience study is novel enough to pursue.

Working title: Fault Lines in a Fly Brain: Mapping Structural Resilience in the Complete Male *Drosophila* Connectome

What the paper does

- Builds a directed, weighted graph from MaleCNS v1.0.
- Traces well-annotated sensory populations to descending and motor populations.
- Measures redundant routes, minimum cuts, and pathway loss under simulated lesions.
- Compares biological wiring with controlled rewired graphs.
- Publishes code, processed graph tables, and an interactive pathway viewer.

What the paper does not claim

- That synapse counts reveal real-time neural activity.
- That a static wiring diagram reproduces learning or behavior.
- That one male specimen represents the full species.
- That a simulated lesion predicts a clinical or biological outcome.
- That structural sex differences alone explain behavior.

The gap this fills

The final Cell article already includes sensorimotor max-flow analysis [1], so basic flow analysis is not a novel contribution. Two 2026 arXiv preprints add relevant prior art: FlyGM reports a connectome-shaped controller for simulated fly locomotion [5], while FLYNN reports robustness experiments in fly-topology robot navigation [6]. The distinct lesion-response and structural-resilience benchmark remains a candidate contribution until a formal literature review establishes novelty.

Novelty gate: Run a dated search across peer-reviewed databases and preprint servers before coding. If an existing study already reports the same MaleCNS whole-CNS lesion benchmark, narrow the study to a modality, such as vision-to-flight control, or to verified dimorphic versus shared circuitry.

Four tests turn the idea into a falsifiable study

ID	Question	Testable hypothesis
RQ1	How redundant are sensory-to-motor routes across annotated modalities?	H1: Route redundancy will differ by modality after controlling for graph size and degree.
RQ2	Which neurons, cell types, and regions produce the largest structural failures when removed?	H2: Removing high-dominator or high-flow-betweenness nodes will reduce reachability more than random or degree-matched removal.
RQ3	Does biological wiring preserve sensorimotor communication better than plausible null graphs?	H3: The observed graph will retain more weighted reachability and disjoint routes than degree-preserving rewires at matched lesion levels.
RQ4	Are dimorphic and male-specific pathways organized around more localized bottlenecks than shared pathways?	H4: Candidate dimorphic subnetworks will show higher concentration of path dependence. This remains exploratory and must be limited to cross-sex mappings verified in the paper or official dataset.

Primary outcome

The main endpoint is the area under the robustness curve for weighted source-to-target reachability as nodes or edges are removed. Each curve will be reported with confidence intervals from repeated perturbation runs.

Secondary outcomes

- Number of vertex-disjoint and edge-disjoint routes
- Minimum vertex and edge cut size
- Weighted shortest-path efficiency
- Communicability between source and target sets

- Fraction of target neurons still reachable
- Critical-node rank stability across metrics
- Effect size against matched null graphs
- Sensitivity to synapse thresholds and edge weights

Decision rule: If H3 is not supported, that is still a useful result. The paper can show that apparent biological robustness is explained by degree distribution rather than a special wiring pattern.

MaleCNS v1.0 is rich enough for the analysis, but one specimen sets a hard boundary

1

fully reconstructed adult male
central nervous system specimen

4

input classes in this plan: neurons,
synapses, annotations, spatial
regions

2

access modes used in this plan:
downloads and programmatic
queries

Required fields

Entity	Required fields	Use in analysis
Neuron	stable body ID, type, class, region, laterality	Node identity, grouping, and stratified analysis
Synapse edge	pre-neuron, post-neuron, synapse count	Directed edge and primary weight
Functional labels	sensory, interneuron, descending, motor	Source, bridge, and target sets
Optional biology	transmitter prediction, fruitless or doublesex label	Signed or dimorphism-aware sensitivity analyses

Initial pathway panels

Start only with pathways whose source and target annotations are clear. The visual panel may use Janelia's official R1-R6 to DNg13 visual-to-motor example as a documented case [4]. It is an anatomical pathway example, not evidence that the static connectome reproduces dynamic behavior. Add olfactory, gustatory, mechanosensory, and proprioceptive panels only after an annotation audit confirms usable coverage.

Scope boundary: The graph represents structural connectivity. Synapse count is a weight proxy, not a measured firing rate. Any dynamic model belongs in a later paper after physiological assumptions can be justified.

A four-stage pipeline keeps every result traceable

1. Ingest

Download approved MaleCNS tables, preserve release metadata, and checksum every raw file.

2. Audit

Check duplicates, orphan IDs, missing labels, edge direction, and synapse-weight distributions.

3. Analyze

Build pathway subgraphs, calculate resilience metrics, and run lesion experiments.

4. Explain

Rank bottlenecks, visualize failure routes, and report uncertainty and sensitivity.

Graph construction

Represent every neuron as a node and each documented directed synaptic connection as an edge. Use summed synapse counts as the base edge weight. Do not add electrical-coupling data unless a separate verified source is available, and never force it into the same directed model. Record each filter in a machine-readable manifest.

Threshold plan

Run the primary analysis at a prespecified minimum synapse count chosen after the distribution audit. Repeat it across at least three thresholds, including the unthresholded graph, to show whether conclusions depend on weak edges.

Pathway definitions

- **Sources:** modality-specific sensory populations with clear annotations.
- **Targets:** descending neurons for brain-to-cord analysis and motor neurons for end-to-end analysis.
- **Allowed paths:** directed paths within a biologically reasonable hop limit, reported as a sensitivity parameter rather than a hidden rule.
- **Compartments:** full CNS, brain-to-descending, and descending-to-motor segments.

Null models

Degree-preserving rewires test whether resilience follows simply from each neuron's number of inputs and outputs.

Region-constrained rewires preserve coarse anatomy while breaking exact neuron-to-neuron structure.

Reproducibility requirement: Fix random seeds, save every graph transform, publish environment files, and make all reported figures regenerable from one command.

Controlled lesions separate important hubs from ordinary high-degree nodes

Experiment	Perturbation	Comparison	Output
E1 Baseline atlas	None	Across modalities	Reachability, path length, disjoint routes
E2 Random failure	Random nodes or edges at rising fractions	1,000 repeated draws per level	Robustness curves and uncertainty
E3 Targeted attack	Degree, betweenness, dominator, and flow ranks	Random and degree-matched removal	Marginal damage and rank stability
E4 Biological groups	Cell type or region removal	Size-matched random groups	Group vulnerability map
E5 Null graphs	Same lesion sets on rewired networks	Observed graph versus null distribution	Standardized effect size and permutation p-value
E6 Sensitivity	Threshold, weight rule, hop limit, uncertain labels	Primary specification	Conclusion stability

Statistical plan

- Report median differences, standardized effect sizes, and 95 percent bootstrap confidence intervals.
- Use permutation tests for observed-versus-null comparisons.
- Control the false discovery rate when testing many cell types or regions.
- Separate confirmatory tests for H1-H3 from exploratory analysis for H4.
- Publish negative and null findings rather than selecting only visually dramatic circuits.

Compute plan: Begin with sparse graph operations and pathway panels. Full all-pairs calculations are unnecessary. Scale only after profiling memory, runtime, and the number of reachable source-target pairs.

Six figures can carry the full argument

Item	What it shows	Why it matters
Fig. 1	Data flow from MaleCNS tables to pathway graphs	Makes preprocessing and exclusions transparent
Fig. 2	Baseline sensory-to-motor redundancy atlas	Establishes which modalities have alternate structural routes
Fig. 3	Random lesion robustness curves	Shows ordinary failure tolerance and uncertainty
Fig. 4	Targeted versus matched lesion curves	Tests whether identified bottlenecks matter beyond degree
Fig. 5	Observed graph versus rewired nulls	Tests whether exact biological wiring adds resilience
Fig. 6	Critical pathway case studies	Turns graph statistics into interpretable neuron and cell-type stories

Tables

- T1 Dataset and filtering summary
- T2 Outcomes by sensory modality
- T3 Top critical nodes, types, and regions with sensitivity results

Supplement

- Full parameter grid
- Annotation audit
- All lesion curves
- Null-model diagnostics
- Reproduction guide

Section-by-section writing plan

1. **Introduction:** Explain why a complete brain-and-cord graph changes the resilience question and state the gap without overstating function.
2. **Results:** Lead with the atlas, then random failure, targeted lesions, null graphs, and case studies.
3. **Discussion:** Separate structural insight from functional interpretation. Address what a single specimen can and cannot support.
4. **Methods:** Make graph rules, thresholds, controls, and statistics complete enough to reproduce.

Novelty and data quality come before model building

Week	Workstream	Exit condition
1	Literature screen and final scope	Novelty matrix confirms the paper does not duplicate FlyGM, FLYNN, or an existing lesion study
2	Data access and provenance	Raw tables download, checksums match, license and citation requirements logged
3-4	Annotation audit and graph build	Source and target sets pass coverage checks; graph statistics reproduce
5-6	Baseline and random lesions	Primary robustness curves run end to end
7-8	Targeted lesions and nulls	Confirmatory tests complete with saved seeds and outputs
9	Sensitivity and case studies	Conclusions remain stable or are narrowed honestly
10	Figures and interactive companion	Every figure regenerates and every pathway links to source IDs
11	Manuscript draft and internal review	Methods, limitations, and claims align
12	Venue verification and submission package	Target venue passes scope, indexing, fee, and integrity checks

Minimum viable paper

One visual pathway, one additional well-annotated modality, three lesion strategies, one degree-preserving null model, a complete sensitivity analysis, and reproducible code. The paper should expand only after this core result is stable.

Roles to secure

Domain advisor

Checks neuron classes, pathway interpretation, and biological wording.

Methods reviewer

Checks graph controls, statistics, and leakage between confirmatory and exploratory work.

Lead analyst

Owns data provenance, code, figures, manuscript, and release package.

Submission rule: Do not pick a journal from reputation alone. Verify current indexing, scope, fees, review model, and predatory signals immediately before submission.

The paper is strongest when its limitations shape the design

Risk	What it means	Design response
Single specimen	Individual variation cannot be estimated from MaleCNS alone.	Frame claims as properties of this mapped connectome and compare only where matched data are valid.
Static structure	Edges do not supply membrane dynamics, timing, neuromodulation, or behavior.	Use structural terms such as reachability and resilience. Do not call the result functional proof.
Weight ambiguity	Synapse count is not the same as physiological influence.	Compare binary, count-weighted, and optional transmitter-aware specifications.
Annotation gaps	Some source or target labels may be incomplete or uncertain.	Publish inclusion rules and repeat analysis after removing uncertain groups.
Scale	Naive all-pairs algorithms may be expensive on 166,700 nodes.	Use sparse matrices, bounded pathway panels, and profiling before full runs.
Novelty overlap	Recent work is appearing quickly.	Freeze the final question only after a dated literature matrix and advisor review.

Go or no-go checkpoints

- **GO IF:** The official release provides stable IDs, synapse edges, and enough functional labels to define at least two pathways.
- **GO IF:** The literature screen finds no study with the same MaleCNS whole-CNS robustness benchmark.
- **NARROW IF:** Labels support only one modality. Publish a deep visual sensorimotor study instead of a broad atlas.
- **STOP IF:** Results change direction under reasonable thresholds or null models and cannot be explained transparently.

Best next action: Complete the novelty matrix and data dictionary first. Those two documents determine whether this becomes a credible paper or just an attractive demo.

Facts, prior work, and proposals are kept separate

- 1. Final peer-reviewed article.** Berg, S., et al. Sexual dimorphism in the complete *Drosophila* male central nervous system connectome. *Cell* 189(18):5504-5526.e15 (2026). Published September 3, 2026. DOI: 10.1016/j.cell.2026.08.015. The final article reports 166,700 neurons and 11,710 neuron types; its Results section reports 124.2 million proofread synaptic connections and a graph of 25.6 million edges between 166,483 neurons. These counts supersede the earlier preprint figures. [Cell full text](#) | [PubMed record](#)
 - 2. Official dataset hub.** HHMI Janelia's Male CNS Connectome Project records the MaleCNS v1.0 release on June 8, 2026 and links the explorers, neuPrint, Clio, downloads, and release notes. The release notes state that the dataset is licensed under CC BY. [MaleCNS project page](#) | [Release notes](#)
 - 3. Official partner account.** Google Research reports more than 166,000 neurons, approximately 125 million synaptic connections, inclusion of the ventral nerve cord, human proofreading, and public visualization and download routes. [Google Research overview](#)
 - 4. Official institutional account.** HHMI Janelia presents R1-R6 visual neurons to DNg13 as a visual-to-motor pathway example. This documents structure, not dynamic behavior. [Janelia overview](#)
 - 5. Preprint, not peer reviewed.** Jin, Z., Zhu, Y., Zhang, C., and Sui, Y. Whole-Brain Connectomic Graph Model Enables Whole-Body Locomotion Control in Fruit Fly. arXiv:2602.17997v3, 2026. It is relevant prior art for connectome-grounded controllers. [FlyGM preprint](#)
 - 6. Preprint, not peer reviewed.** Wang, B. and Chen, J. FLYNN: Robust Neural Network for Robot Navigation using Fly Brain Topology. arXiv:2607.00025v1, 2026. It is relevant prior art for topology-based robustness claims. [FLYNN preprint](#)
 - 7. Peer-reviewed methods paper.** Wang-Chen, S., et al. NeuroMechFly v2: simulating embodied sensorimotor control in adult *Drosophila*. *Nature Methods* 21(12):2353-2362 (2024). DOI: 10.1038/s41592-024-02497-y. [PubMed record](#)
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Integrity rule for the build: Dataset counts and biological descriptions must trace to sources 1-4. Sources 5-6 establish prior art only and are labeled as preprints. Hypotheses, thresholds, lesion counts, architecture choices, and the 12-week schedule are proposed methods, not reported findings. Before submission, verify the dataset license and required acknowledgments from the official release records.